

Evolution and strengthening of nanoprecipitates in a high strength maraging stainless steel

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ABSTRACT

A high strength maraging stainless steel (MSS) was developed utilizing multiple nanoprecipitates and reverted austenite. After aging treatment, this steel shows remarkable mechanical properties with a yield strength of ~ 1750 MPa, a tensile strength of ~ 1910 MPa, and a total elongation of $\sim 10.5\%$. The precipitation and strengthening mechanisms of multiple nanoprecipitates were clarified. The precipitation sequence upon aging can be identified as: Ni-rich cluster $\rightarrow$ Ni-rich cluster + Mo-rich cluster $\rightarrow$ Ni_3Ti + Mo-rich phase $\rightarrow$ Ni_3Ti + Mo-rich phase + α' -Cr. Ni-rich cluster and Mo-rich cluster is the precursor of Ni_3Ti and Mo-rich phase, respectively. The Ni-rich clusters, Mo-rich clusters, Ni_3Ti and α' -Cr, contribute to the precipitation strengthening mainly by shearing mechanisms, among them the ordered strengthening plays the most significant role. Upon aging, Ni_3Ti is wrapped by Mo-rich phase, restraining their growth. In contrast, the Mo-rich phase can be significantly coarsened, leading to a shift in the precipitation strengthening from dislocation shearing to Orowan looping mechanism. The reverted austenite improves the ductility of the MSS via a transformation induced plasticity (TRIP) effect.

1. Introduction

Maraging stainless steels (MSSs) with high strength, ductility and corrosion resistance are the backbone of innovations in manufacturing, energy, transportation, and safety, such as automotive engine housings, aircraft landing gears, and cryogenic missiles [1]. These excellent properties stem from the formation of a unique microstructure during the aging treatment: hierarchically arranged lath martensite with a high dislocation density and decorated by various nanoprecipitates and reverted austenite (RA) that re-precipitates from martensite [2,3]. The high dislocation density of the martensite matrix and the multiple nanoprecipitates provide the high strength of MSSs [2,4]. The RA contributes to the plasticity and toughness [5,6]. However, the complex interactions between alloying elements and the varying microstructure during aging tend to be difficult to achieve an excellent balance between strength and ductility in MSSs. In particular, coarse nanoprecipitates or severe grain boundary segregation of atoms can lead to brittle fracture

[7,8]. Therefore, it is essential to deeply investigate the precipitation mechanism of nanoprecipitates during aging treatment.

Much attentions have been paid to the relationship between nanoprecipitates and mechanical properties, while the formation mechanism of nanoprecipitates during aging treatment is not well understood [9–11]. In MSSs, three types of nanoprecipitates can be formed during aging treatment, including Mo-rich, Ni_3Ti , and Cr-rich nanoprecipitates, which significantly increased the strength [10]. The effect of alloying elements on the precipitation precipitation-hardened steels was investigated by transmission electron microscopy (TEM), atom probe tomography (APT) and first-principles calculations [12,13]. Tian et al. found that the addition of Co can provide stronger thermodynamic driving force to form Ni-Ti clusters in body-centered cubic (BCC) iron and promote the formation of Ni_3Ti [12]. The effect of multiple alloying additions on the co-precipitation of Mo-rich, Ni_3Ti and Cr-rich nanoprecipitates was investigated by Niu et al. [13]. They found that Mo could enhance the chemical driving force and reduce the strain energy of

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nucleation of Ni₃Ti, which promotes the formation of Ni₃Ti. While Ti and Mo could consume Ni in the matrix, which reduces the tendency of spinodal decomposition for Cr and suppresses the precipitation of α' -Cr. However, many studies usually neglect the investigation of the precipitation sequence mechanism during aging, which can significantly affect the strength of the steels. The evolution mechanism of nanoprecipitates in precipitation hardening steels through aging treatment process has an important effect on the mechanical properties [12,14–16]. For example, in maraging steels utilizing Cu and NiAl as nanoprecipitates, the ratio of Ni and Al influenced the precipitation sequence during aging treatment, resulting in significantly different mechanical properties [17,18]. Exceeding the aging peak, the precipitates begin to coarsen, and Cu-rich precipitates with a 9R structure convert to face-centered cubic (FCC) structures [19], and the NiAl-type precipitates gradually convert to Ni₂AlMn-type precipitates [20]. However, the precipitation sequence of Mo-rich, Ni₃Ti, and Cr-rich nanoprecipitates in maraging stainless steels during aging is still not clear and requires further investigation to better understand the strengthening mechanism.

Although precipitation strengthening can significantly increase the strength of steels, it often sacrifices ductility to a certain extent [14]. The phenomenon is a trade-off between strength and ductility [21–24]. To overcome this issue, researchers have tried to introduce the transformation induced plasticity (TRIP) effect of austenite to improve ductility [5,25,26]. In maraging steels, the aging treatment is also accompanied by RA re-precipitation [26,27]. Different insights are available on the formation mechanism of RA. The segregation of austenite stabilizing elements at grain boundaries, such as Ni and Mn, can reduce the Gibbs free energy of austenite and promote the formation of RA [28–30]. Heo et al. suggested that the RA was formed by the coherent strain at the interface between Ni₃Ti and the matrix as a driving force through the climbing of the edge dislocations and the diffusion of Fe atoms [31]. Typically, the RA in MSS can effectively improve ductility [25,26]. However, how to introduce and better control the formation of nanoprecipitates in RA remains unknown and needs to be further explored.

In the present work, an ultra-high strength MSS was developed utilizing multiple nanoprecipitates and RA. Microstructures at different length scales were characterized by electron backscatter diffraction (EBSD), TEM and APT. First principles calculations were also conducted to explain the mechanism of nanoprecipitates formation. The objective for this work is to examine the effect of aging on microstructural evolution and mechanical properties in order to elucidate the mechanisms of nanoprecipitation strengthening. Meanwhile, the mechanism of nanoprecipitate formation during aging treatment is also elaborated.

2. Experimental methods

The MSS with a nominal composition of Fe-11.0Cr-4.0Co-8.0Ni-0.5Ti-5.0Mo-0.1Si-0.002C (wt.%) was designed and fabricated. Alloy rods, weighing approximately 500 g, were fabricated from constituent elements (>99.95 at.% purity) by arc melting in an argon atmosphere on a water-cooled copper vessel. Prior to alloy melting, the furnace chamber was evacuated, flushed and backfilled with high purity argon. The alloy rods were remelted at least five times to ensure the homogeneity and dropped into a copper mold with dimensions of Φ 20 mm $\times$ 200 mm.

The as-cast ingots were hot-rolled at 1200 °C from 20 to 2 mm in thickness, followed by water quenching. The specimens were solid solution treated at 1050 °C for 1 h and quenched in ice-water mixture, followed by cryogenic treatment in liquid nitrogen for 8 h in order to obtain single-phase martensite (labeled as ST). The sheets were then aged at 480 °C for 15, 30 (labeled as SA30), 120, 300 (labeled as SA300), 3,000, 12,000 and 30,000 min (labeled as SA30000), respectively.

Hardness measurements were conducted using a Rockwell tester with a load of 500 g for 15 s. At least ten indentations were measured to obtain an average value. Tensile tests were performed with an Instron

5567 testing machine along the rolling direction of the samples with a strain rate of 10^{-3} s⁻¹. Three specimens were tested in each case to ensure the repeatability. Dimensions of the flat dog bone shaped tensile tests are 12.5 mm, 3.8 mm and 2.0 mm in length, width and thickness, respectively. To clean up the scratches, the surfaces of the test samples were polished with 3000-grit SiC paper. A contacting Instron extensometer was used to measure the strain within the sample gauge upon loading. The yield strength was determined using the 0.2 % offset plastic strain method.

X-ray diffraction (XRD) measurements were carried out using a D/max-2550 X-ray diffractometer to characterize the phase composition. The instrument operated at 40 kV and 40 mA, utilizing Cu - K α radiation with a sampling interval of 0.01° and a scanning rate of 1°/min. Volume fractions of the dual phase were calculated by the results of XRD. The detailed method has been reported in a previous work [4]. The dislocation density (ρ) was estimated from the XRD profile by Williamson Hall method as Eqs. (1) and (2) [32]:

$$\frac{\Delta 2\theta \cos \theta}{\lambda} = \frac{0.9}{D} + 2\varepsilon \frac{\sin \theta}{\lambda} \quad (1)$$

$$\rho = \frac{2\sqrt{3}\varepsilon}{Db} \quad (2)$$

where λ is the X-ray wave length ($\lambda = 0.154$ nm), D is an apparent crystallite size, and ε is a lattice strain. b is the magnitude of the Burgers vector, which is 0.248 nm for iron [33]. The four diffraction peaks of martensite were record for (110), (200), (211) and (220). To accurately and quantitatively evaluate the dislocation density by XRD, the reference samples were subjected to stress relief by holding at 1050 °C for 20 h.

EBSD measurements were carried out using a SEM (Hitachi S3400) equipped with an HKL Channel 5 software. EBSD specimens were mechanically polished and then ion-beam polished.

TEM observations and selected area electron diffraction (SAED) were conducted on a JEM 2100F microscope operated at 200 kV. High-angle annular dark field scanning transmission electron microscope (HAADF-STEM) and STEM-energy dispersive X-ray (STEM-EDX) observations were carried out on an aberration-corrected Titan G260-300 microscope operated at 300 kV. Thin TEM/STEM foils were prepared by lift-outs in an FEI Scios focused ion beam (FIB) equipped in SEM.

Needle-shaped specimens for APT were fabricated by lift-outs and were annular milled using FIB [34]. APT characterization was performed in a local electrode atom probe (CAMECA LEAP 5000 XR). Specimens were analyzed in voltage mode at 70 K with a pulse repetition rate of 200 kHz, a pulse fraction of 20 %, and an atomic evaporation detection rate of 0.3 % per pulse. The Imago Visualization and Analysis Software (IVAS) version 3.8 was used to create 3D reconstructions and data analysis.

Through first principles calculations, atom occupancy and formation energies of multicomponent nanoprecipitates with different structures were investigated using the Cambridge Sequential Total Energy Package (CASTEP) module of the Materials Studio 8.0 software package with the generalized gradient approximation (GGA) function of Perdew-Burke-Ernzerh (PBE). To make the formation energies comparable, BCC structures were constructed to determine the total energy of each cell. The cut-off plane-wave energies of BCC Fe was fixed at 320 eV. The k-point lattices of the 2-atom BCC Fe as $2 \times 2 \times 2$. The calculations considered the magnetic properties. Meanwhile, both the ionic and lattice parameters were allowed to be fully relaxed.

3. Result

3.1. Mechanical properties

Fig. 1 (a) shows the variation of hardness with aging time for the

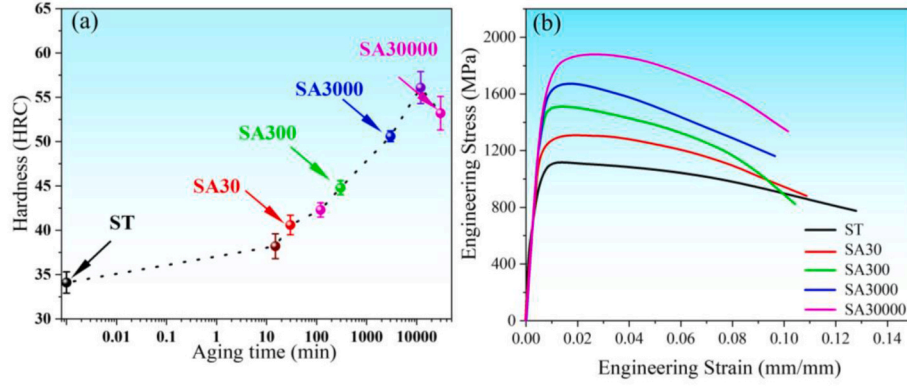


Fig. 1. Mechanical properties for MSS aged at 480 °C for various times. (a) The hardness evolution of MSS, and (b) engineering stress-strain curve of MSS, where ST stands for solution-treated MSS.

developed MSS. The hardness increases from ~34.1 to ~48.2 HRC within the initial 15 min and gradually reaches a maximum of ~56.1 HRC at 12,000 min. Subsequently, the hardness begins to decrease with further increase in aging time owing to the over-aging effect.

Ambient temperature tensile tests were carried out to further study the mechanical properties after the solution and aging treatments. The engineering stress–strain curves are presented in Fig. 1 (b). ST MSS

shows a yield strength (YS) of 850 MPa, a tensile strength (TS) of 1100 MPa and a total elongation (TE) of 12.4 %. After aging for 300 min, the YS increases to 1350 MPa and TS reaches 1510 MPa. Extending the aging time to 30,000 min, the YS increases to 1750 MPa, and the TS reaches up to 1910 MPa. At the same time, there is no significant compromise in TE. Notably, the MSS retains high strength and considerable plasticity even in over-aged conditions.

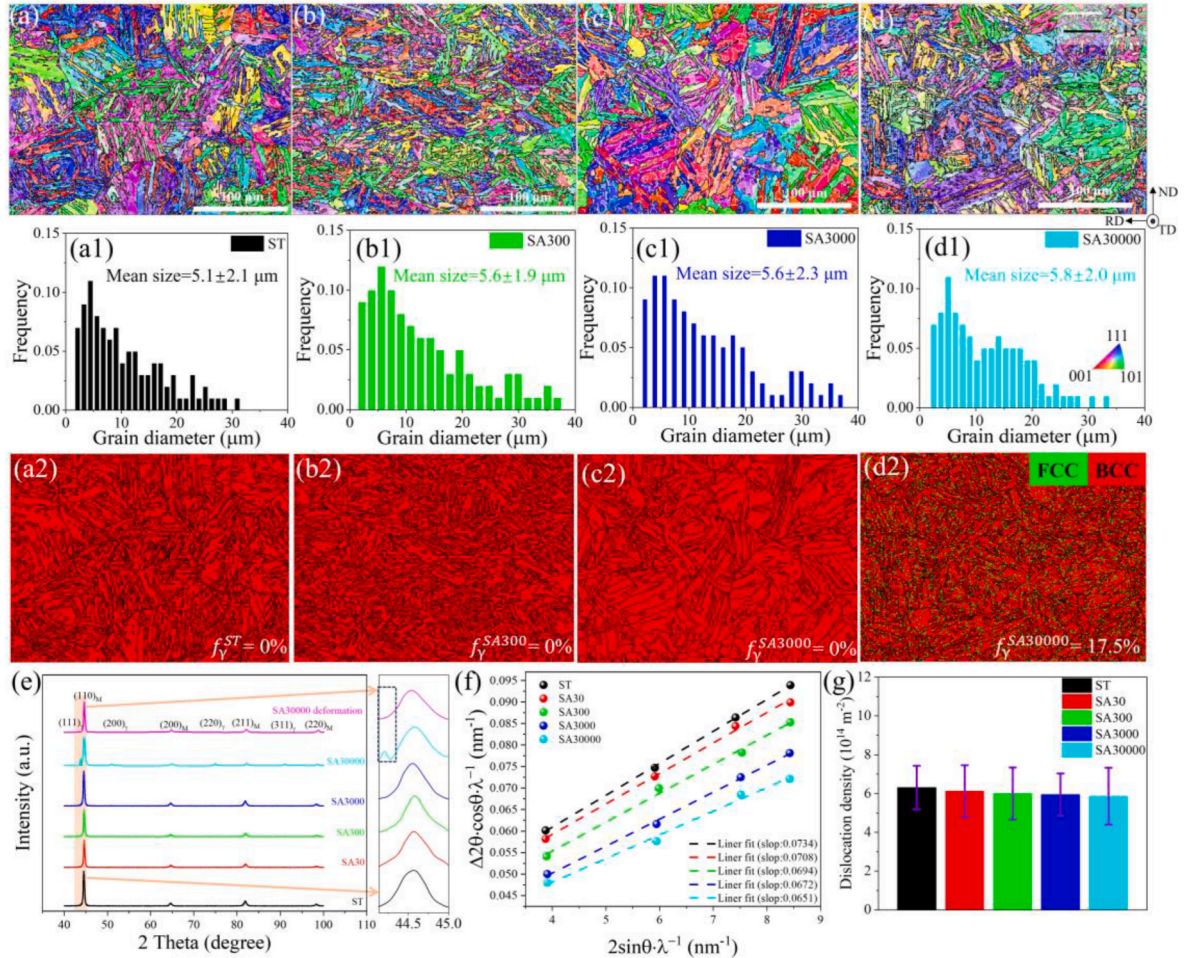


Fig. 2. Characterizations of the matrix microstructure at different aging states. Orientation mappings, equivalent effective grain size distribution and phase distribution maps of (a), (a1) and (a2) for ST MSS, (b), (b1) and (b2) for SA300 MSS, (c), (c1) and (c2) for SA3000 MSS and (d), (d1) and (d2) for SA30000 MSS. (e) XRD patterns of ST, SA30, SA300, SA3000, SA30000 MSS and SA30000 MSS vicinity of the fracture. (f) The Williamson-Hall curve of martensite. (g) Dislocation density histograms for different aging states.

3.2. Matrix microstructural characterization

Fig. 2 displays the microstructure of the matrix at different aging states. The ST, SA300, SA3000 and SA30000 MSSs possess a typical martensite hierarchical microstructure. The orientation mappings with low ($2\text{--}15^\circ$) and high angle ($>15^\circ$) grain boundaries (HAGBs) of ST, SA300 and SA30000 MSS obtained by EBSD are shown in Fig. 2(a–d). HAGBs can serve as boundaries separating crystalline regions to form equivalent effective grains [35,36]. Fig. 2 (a1–d1) display the statistics distribution of martensite equivalent effective grain dimensions. The mean equivalent effective grain dimensions of the ST, SA300, SA3000 and SA30000 MSS are ~ 5.1 , ~ 5.6 , ~ 5.6 and $\sim 5.8\ \mu\text{m}$, respectively. It is found that no significant change in grain dimensions was observed before and after aging. Fig. 2 (a2–d2) show the phase distribution map for the MSSs. The matrix of the ST, SA30, SA300 and SA3000 MSS is composed fully of martensite. The matrix of the SA30000 MSS consists of dual martensite and austenite phases, where the volume fraction of austenite is 17.5 %. The austenite distributes homogeneously in the matrix.

The XRD results show that in the ST, SA30, SA300 and SA3000 MSSs, the matrix consists of single-phase martensite, as shown in Fig. 2 (e). However, the matrix of SA30000 MSS consists of dual martensite and austenite phases, where the volume fraction of austenite is $15.5 \pm 3.2\%$. It is consistent with the EBSD observation. XRD results show that no austenite peaks can be detected before aging for 3000 min. Only after aging for 30,000 min can the austenite be detected, indicating that the formation of RA is a slow process and depending on aging time. As no austenite is observed after ST (i.e., residual austenite), this austenite can be identified as RA, which is very common in maraging steels [37]. After deformation, the austenite (111) peak in the XRD pattern from SA30000 MSS disappeared (Fig. 2 e), indicating that the RA transformed to martensite. This suggests that the TRIP effect occurred during deformation. Fig. 2 (f) shows Williamson-Hall plots. With increasing aging time, the slope of fitted lines becomes smaller. Fig. 2 (g) demonstrates the dislocation density of MSS in different aged conditions. The dislocation density of MSS is ~ 6.21 , ~ 6.13 , ~ 6.01 , ~ 5.95 and $\sim 5.72 \times 10^{14}\ \text{m}^{-2}$, respectively, which is consistent with that of maraging steels [38,39]. This implies that aging treatment does not affect the dislocation density of the matrix.

Fig. 3 (a) and (a1) demonstrate that the matrix comprises martensite laths with a high dislocation density in the ST MSS. No additional reflections are detected in the SAED pattern along the $[001]_M$ direction, indicating that no nanoprecipitate exists in this state. After aging for 300 min, the matrix remains lath martensite with a high dislocation density, and nanoprecipitates are found in the SAED pattern along the $[011]_M$ direction, as shown in Fig. 3(b), Fig. 3 (b1) and Fig. 3 (b2). Simultaneously, rod-like Ni_3Ti and ellipsoidal Mo-rich phase were observed in the lath. With increasing aging time, rod-like Ni_3Ti and ellipsoidal Mo-rich phase grow up, as shown in Fig. 3(c)–Fig. 3 (c1) and Fig. 3 (c2). These rod-shaped Ni_3Ti precipitates are generally smaller than 10 nm in length and 3 nm in width. Meanwhile, film-like RAs along the lath boundary and blocky RAs are also observed. Fig. 3 (c3) demonstrates that the RA and matrix possess an N-W orientation relationship: $(111)_\gamma // (011)_M$, and $[\bar{1}\bar{1}0]_\gamma // [\bar{1}00]_M$. In MSS, it is common for both the austenite and martensite to exhibit K-S and N-W orientation relationships [1]. Fig. 4 demonstrates that no elements are segregated in RA. This indicates that RA does not contain nanoprecipitates. HRTEM of the nanoprecipitates is shown in Fig. 5(a) and (b). The fast Fourier transform (FFT) patterns of Fig. 5 (a1) and (b1) demonstrate the nanoprecipitates possess the $\eta\text{-Ni}_3\text{Ti}$ structure. The orientation relationship between the Ni_3Ti phase and martensite matrix is $(0004)_\eta // (110)_M$ and $[1\bar{2}10]_\eta // [001]_M$. The lattice mismatch between the matrix and $\eta\text{-Ni}_3\text{Ti}$ phase can be described as $\delta = \frac{|a_p - 2a_m|}{2a_p + a_m}$, where α refers to the lattice parameter of each phase [40]. The δ is evaluated to be about 0.03 and 0.04 based on the HRTEM results. Fig. 5 (a2) and (b2) demonstrates the

Mo-rich phase with a hexagonal structure, which agrees with the results in the literatures [41]. The orientation relationship between the Mo-rich phase and martensite matrix is $(0001)_{\text{Mo}} // (01\bar{1})_M$ and $[01\bar{1}0]_{\text{Mo}} // [\bar{1}11]_M$. The δ is evaluated to be about 0.11 and 0.12 based on the HRTEM results. In MSS, the spinodal decomposition of Cr forms the $\alpha'\text{-Cr}$ phase [42]. The $\alpha'\text{-Cr}$ phase usually possesses a BCC structure and remains coherent with the matrix [13].

3.3. Characterization of nanoprecipitates

APT was employed to gain further characterization for the evolution of the various precipitates. All atoms are uniformly distributed in the matrix after ST, as shown in Fig. 6 (a). Aging for 30 min, the enrichment of Ni and Ti is observed and Ni-rich cluster are formed, as shown in Fig. 6 (b). These Ni-rich cluster are the precursor phase of Ni_3Ti during the aging process [12]. When the aging time is extended to 300 min, the enrichment of Mo is found in Fig. 6 (c), forming Mo-rich clusters. Further aging to 3000 min, the Ni-rich and Mo-rich clusters gradually grow into nanoprecipitates, as shown in Fig. 6 (d). Interestingly, after aging at 480°C for 30,000 min, further enrichment of Cr is observed with the formation of $\alpha'\text{-Cr}$, as shown in Fig. 6 (e). In MSS, the spinodal decomposition of Cr forms the $\alpha'\text{-Cr}$ [42]. The $\alpha'\text{-Cr}$ phase usually possesses a BCC structure and remains coherent with the matrix [13].

The evolution of the nanoprecipitates were observed by APT at different aging conditions, as shown Fig. 7(a–d). In the early stage of aging, only Ni-rich clusters are present, while Mo-rich and $\alpha'\text{-Cr}$ phases gradually appear with the increasing aging time. Fig. 7 (a1–d3) show the change in the center composition of the nanoprecipitates and matrix at different aging conditions. Ultimately, the Mo-rich and $\alpha'\text{-Cr}$ phase coarsen, as shown in Fig. 7, which is consistent with the previous studies [14,43,44]. It can be found that the concentrations of Ni and Ti in Ni-rich cluster cores gradually increase, while the concentrations of Fe, Co, Mo and Cr gradually decrease. Finally, the ratio of Ni to Ti in the cluster cores approaches 3:1 and reaches a stable state, forming the hexagonal structure Ni_3Ti . Typically, MSS with a high dislocation density after solution treatment leads to the formation of Ni-rich cluster by the diffusion of atoms along the dislocation lines during the early stages of aging [33,45]. Fig. 7 (a) displays that Ni-rich cluster are precipitated uniformly with a considerable number density. The matrix possessing high dislocation density allows for the prevalence of continuous precipitation in the matrix. During plastic deformation, additional energy from the anti-phase boundary is demanded by the dislocations shearing these nanoprecipitates, which is the major factor for increasing the strength at the early aging [45].

The Ni-rich cluster formed at the beginning of aging can serve as heterogeneous nucleation sites for the Mo-rich and $\alpha'\text{-Cr}$ phase. As the aging time increasing, some of the Mo-rich and $\alpha'\text{-Cr}$ phase form adjacently to the Ni-rich cluster, as shown in Fig. 7(a–d). At the same time, the exclusion of the core atoms of Ni-rich cluster also promotes the heterogeneous nucleation of Mo-rich and $\alpha'\text{-Cr}$ phase. The nanoprecipitates possessing different shapes can be approximately transformed into spheres of identical volume phases with an equivalent radius [46]. The volume fraction of the nanoprecipitates was estimated by $f = 4/3\pi r^3 N$, where r and N are the nanoprecipitates radius and number density, respectively. Table 1 shows the equivalent radius and number density of the nanoprecipitates at different aging conditions. Increasing the aging time from 30 to 3000 min increased the size of the Ni-rich cluster from 1.21 to 3.79 nm. While from 3000 to 30,000 min, the Ni_3Ti did not coarsen significantly. Because the Mo-rich phase wrapped around Ni_3Ti , which inhibits the coarsening of Ni_3Ti (Fig. 7b–d). The above-mentioned Ni-rich and Mo-rich clusters are the precursor of Ni_3Ti and Mo-rich phases, respectively. The chemical compositions and equivalent radii of the Ni-rich clusters and Mo-rich clusters are 30.7Fe-5.6Co-38.5Ni-10.7Ti-5.3Mo-8.8Cr-0.4Si with 0.95 nm and 51.6Fe-2.2Co-11.5Ni-0.6Ti-21.2Mo-10.8Cr-2.1Si with 1.47 nm,

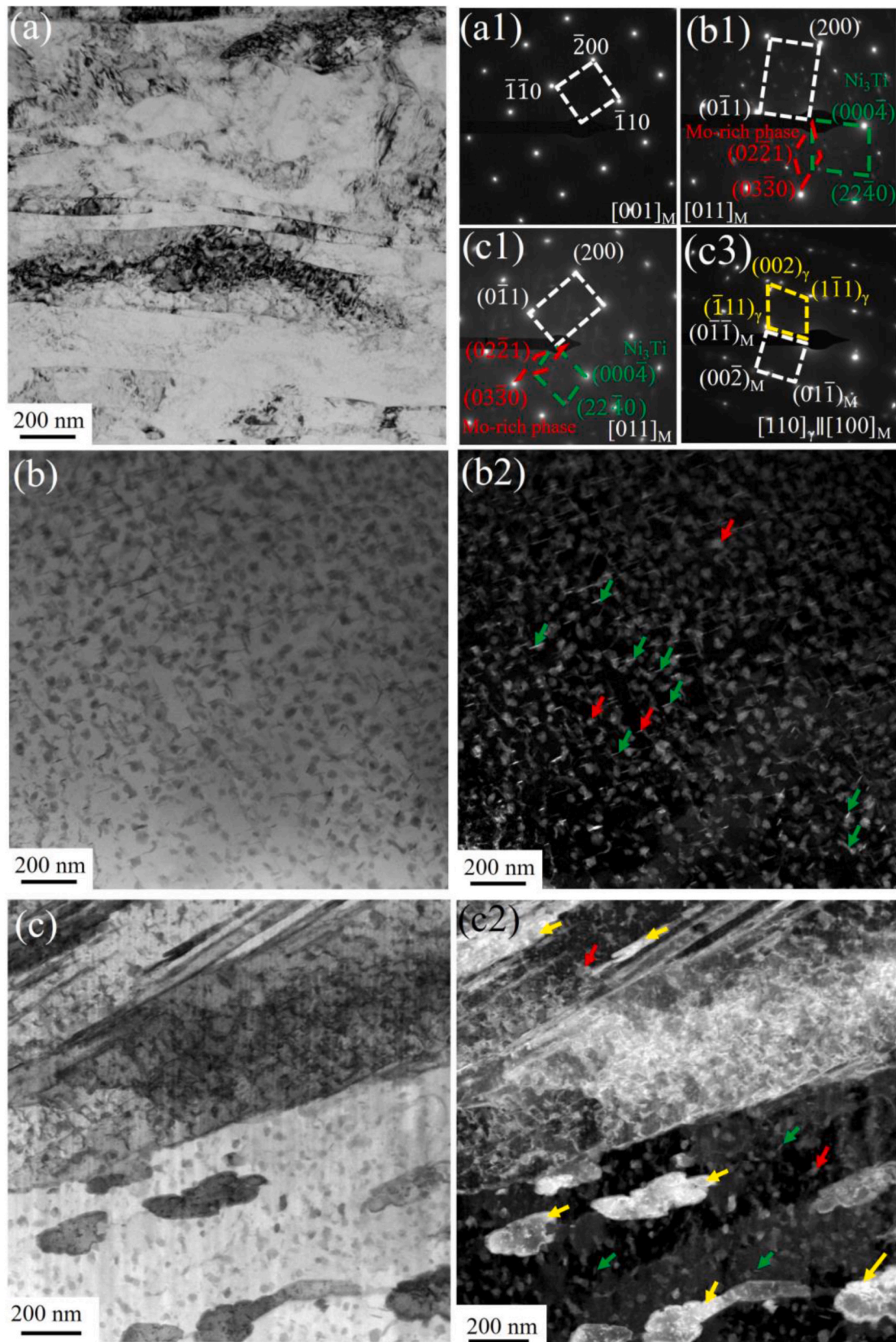


Fig. 3. Bright-field TEM images of (a) ST MSS, (b) SA300 MSS and (c) SA30000 MSS. (a1), (b1), (c1) and (c3) correspond to SAED in the figures. Dark-field TEM images of (b2) SA300 MSS and (c2) SA30000 MSS. Green denote Ni_3Ti , red denote Mo-rich phase and yellow arrows denote RA. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

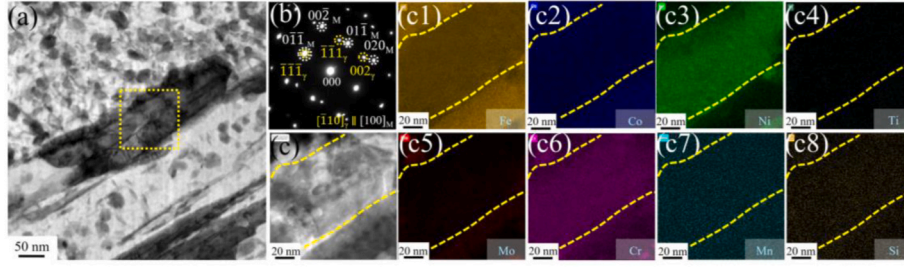


Fig. 4. (a) BF-TEM of RA in SA30000 MSS. (b) SAED of RA and matrix, (c) STEM-HAADF image of yellow dashed frame in (a), (c1-c8) STEM-EDX elemental distribution maps. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

respectively. Due to the high dislocation density in the quenched matrix, clusters are formed by diffusion along the dislocation lines during aging [47]. High number density of Ni-rich and Mo rich clusters form uniformly, coherent with the matrix. The clusters grow up to form Ni_3Ti with D_{024} structure and Mo-rich phase with hexagonal structure. Their compositions are $1.4\text{Fe}-1.0\text{Co}-72.4\text{Ni}-23.8\text{Ti}-0.3\text{Mo}-1.0\text{Cr}-0.1\text{Si}$ and $33.5\text{Fe}-1.2\text{Co}-13.6\text{Ni}-0.3\text{Ti}-28.9\text{Mo}-18.9\text{Cr}-3.6\text{Si}$ with sizes 3.89 nm and 8.02 nm, respectively. These nanoprecipitates effectively improve the strength of the matrix through the precipitate shearing or Orowan looping mechanism. The change in precipitation sequence with increasing aging time is the primary contributor for the variation in strength increment.

4. Discussions

4.1. Precipitation mechanism

The APT results after solution treatment showed that Fe, Co, Ni, Ti, Mo, Cr, and Si were homogeneously distributed without any enrichment (Fig. 6). However, after aging at 480 °C, Ni, Ti, Mo, Si and Cr atoms become significantly enriched, forming Ni_3Ti , Mo-rich and α' -Cr phases, respectively (Fig. 6). The appearance of these nanoprecipitates follows a certain precipitation sequence. Initially, Ni-rich clusters are formed. The Mo-rich clusters appear with the increased aging time, and these Mo-rich clusters prefer heterogeneous nucleation at interfaces between the Ni-rich cluster and matrix. That is consistent with the findings for others [10,45,48]. The reason is that Ni, Ti and Mo possess different diffusion rates in α -Fe [49–52]. Meanwhile, the Ni-Ti precipitates cores expel Fe, Co, Mo and Cr atoms and gradually form a stable Ni_3Ti phase (Fig. 7). With further prolonged aging time, the α' -Cr phase nucleates and grows. The matrix shows the coexistence of Ni_3Ti , Mo-rich and α' -Cr phases. It is the precipitation sequence of the nanoprecipitates that leads to different precipitation strengthening effects at different aging stages (Fig. 1).

Studies on the precipitation mechanism of MSS tend to focus on the characteristics of the nanoprecipitates at peak aging, while ignoring the sequence of the nanoprecipitates [53–55]. The nanoprecipitates at different aging time plays a very important role in controlling the mechanical properties [55,56]. It is reported that the effect of Co on the strength in maraging steels is extremely important [12]. The addition of Co promotes the nucleation of nanoprecipitates and further increases the number density of nanoprecipitates, thus enhancing the precipitation hardening effect [42,57]. The results of APT show a random distribution of Co atoms in the matrix (Fig. 6). Therefore, Fe-Co-based alloys with randomly distributed Co were used in the first-principles calculation.

To explore the precipitation sequence of MSS, first-principles calculation was utilized to investigate the distribution of different contents of X (Ni, Mo or Cr atoms) in Fe-Co-X alloys. Fig. 8 shows two tendencies of X distribution in the Fe-Co-X alloy system, forming precipitates (Fig. 8 a) and dispersed distribution (Fig. 8 b). Table 2 shows the calculated details of the alternative atoms X (Ni, Mo and Cr) preferring cluster or disperse distribution in different alloy systems. M-I shows the distribution tendency for two substitution atoms, M-II and M-

III for three substitution atoms, M – IV and M – V for five substitution atoms and M – VI for nine substitution atoms.

The tendency of different substitutional atom X can be expressed by Eq. (3):

$$\Delta E_{\text{cluster/disperse}}^{X-M} = \left(E_{(M,128-M)}^{\text{TotalX}} - M \times \mu_X - (128 - M) \times \mu_{\text{Fe}} \right) / 128 \quad (3)$$

where $E_{(M,128-M)}^{\text{TotalX}}$ represents the total energy of the supercell with M substitutional atoms (X). μ_X and μ_{Fe} are the chemical potentials of the substitutional atoms and Fe, respectively, which were computed by presuming the identical cell symmetry.

To identify whether X in Fe-Co-X alloys prefers to be dispersed or clustered in the system, Fig. 8 (c) shows the cluster formation energy versus the X concentration. Fig. 8 (c) clearly shows that in the alloy system of Fe-Co-X, the evaluated ΔE reduces with the increasing concentration of the substituted atoms, regardless of the distribution the atoms.

For Ni, when there are two atoms in the system, $\Delta E_{\text{disperse}}^{\text{Ni}-2} < \Delta E_{\text{cluster}}^{\text{Ni}-2}$, which means that the two Ni atoms tend to be dispersedly distributed. When the number of Ni atoms in the system increases to three, $\Delta E_{\text{disperse}}^{\text{Ni}-3} > \Delta E_{\text{cluster}}^{\text{Ni}-3}$, this means that three Ni atoms tend to gather into a cluster. With the increasing number of Ni atoms in the system, the formation energy of dispersed distribution is always larger than that of cluster distribution, which means that Ni atoms prefer to be clustered when the number of Ni atoms in the system is higher than three. For Mo, when the system contains two or three Mo atoms, $\Delta E_{\text{disperse}}^{\text{Mo}-2} < \Delta E_{\text{cluster}}^{\text{Mo}-2}$ and $\Delta E_{\text{disperse}}^{\text{Mo}-3} < \Delta E_{\text{cluster}}^{\text{Mo}-3}$, which indicates that Mo tends to be dispersedly distributed. When the number of Mo atoms in the system increases to five, $\Delta E_{\text{disperse}}^{\text{Mo}-5} > \Delta E_{\text{cluster}}^{\text{Mo}-5}$. In this case, the Mo atoms prefer to be distributed as a cluster. In the case of Cr, until the number of Cr atoms in the system reaches 9, $\Delta E_{\text{disperse}}^{\text{Cr}-9} > \Delta E_{\text{cluster}}^{\text{Cr}-9}$, suggesting that chromium atoms are more inclined to form clusters. This result coincides well to the miscibility gap in the α -phase for Fe-Cr-Co alloys [12]. This also explains why multiple nanoprecipitates appear in this alloy system. These types of nanoprecipitates are also present in other Ni-Mo-Cr-Ti system MSSs and effective in enhancing the precipitation strengthening effects [12].

At the same time, comparing the minimum number of atoms required for cluster distribution and the formation energy, the system requires 3 atoms for Ni, 5 atoms for Mo and 9 atoms for Cr for forming clusters, and $\Delta E_{\text{cluster}}^{\text{Ni}-3} < \Delta E_{\text{cluster}}^{\text{Mo}-5} < \Delta E_{\text{cluster}}^{\text{Cr}-9}$. It can be found that Ni requires the lowest number of atoms and the lowest formation energy when distributed in clusters, while Cr requires the highest number of atoms and the highest formation energy. This leads to a certain precipitation sequence during aging, with Ni-rich clusters precipitation first, then Mo-rich nanoprecipitates, and finally Cr-rich nanoprecipitates.

The number of atoms in the cluster and the interatomic distance affect the stability of the cluster structure. Bake et al. found that increasing the number of atoms in the cluster raises the stability of the cluster and facilitates the formation of clusters [58]. Kong et al. found that smaller interatomic distance in the clusters improves the stability of

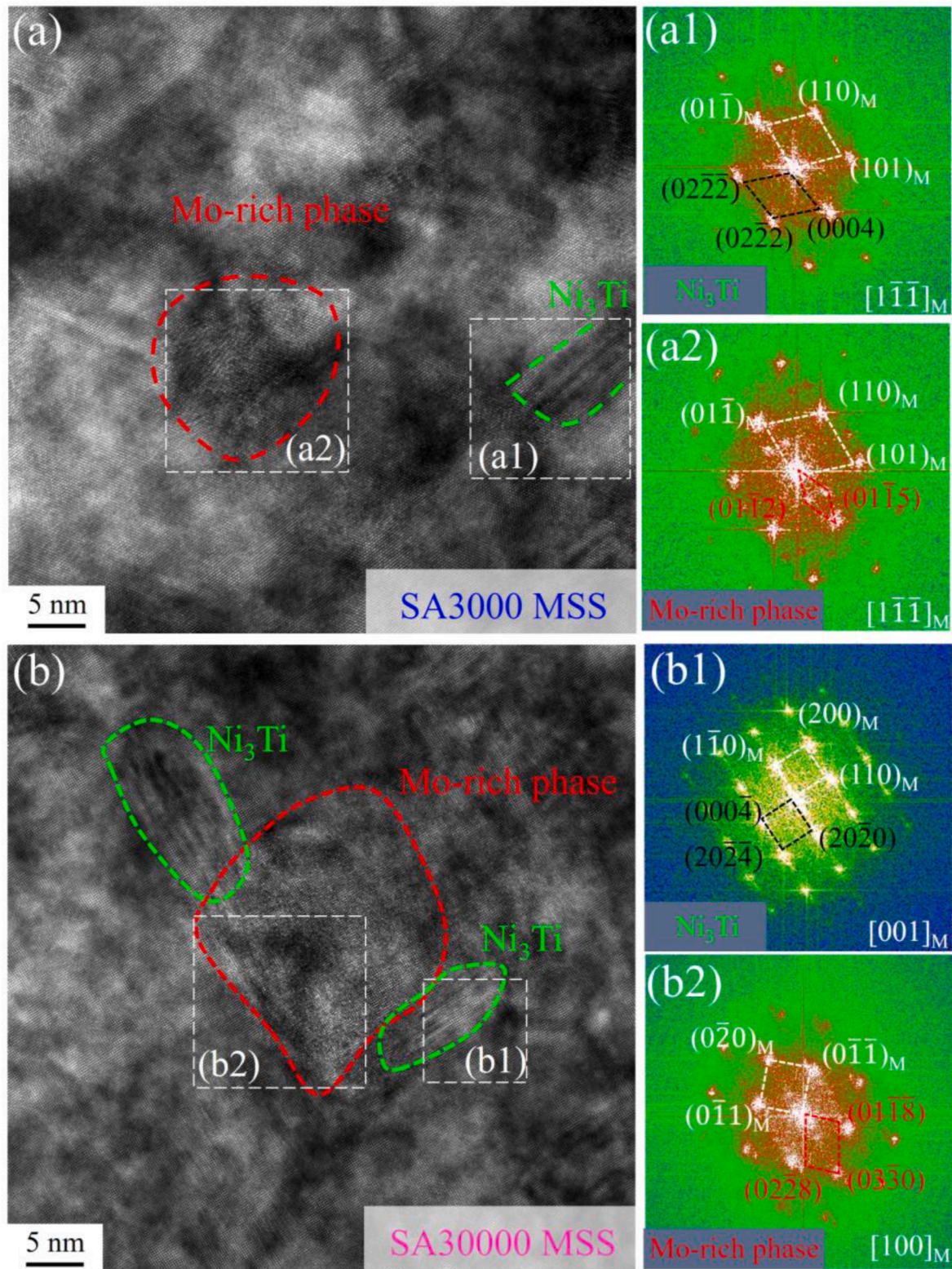


Fig. 5. HRTEM of (a) SA3000 MSS and (b) SA30000 MSS. FFT of (a1) Ni₃Ti in SA3000 MSS, (a2) Mo-rich phase in SA3000 MSS, (b1) Ni₃Ti in SA30000 MSS and (b2) Mo-rich phase in SA30000 MSS. White dashed frames denote the locations of FFT.

the clusters [59]. In the M – IV model, the average atom distance of the clusters (2.47 Å) is lower than that of the dispersed model (4.9 Å). This will increase the stability of the clusters, resulting in a lower formation energy of the cluster structure.

At the beginning of aging, no Mo-rich and Cr-rich nanoprecipitates are found due to the high formation energy for clusters and the low

diffusion coefficient of Mo and Cr [49–51]. Therefore, only Ni-rich cluster is formed, as shown in Fig. 7 (a). The atomic ratio of Ni to Ti in the Ni-rich cluster gradually approaches 3:1, and reaches a stable condition to form Ni₃Ti. The formation energy of cluster distribution for Mo atoms is lower than that of Cr atoms. At this time, the interface between Ni₃Ti and the matrix acts as a heterogeneous nucleation site.

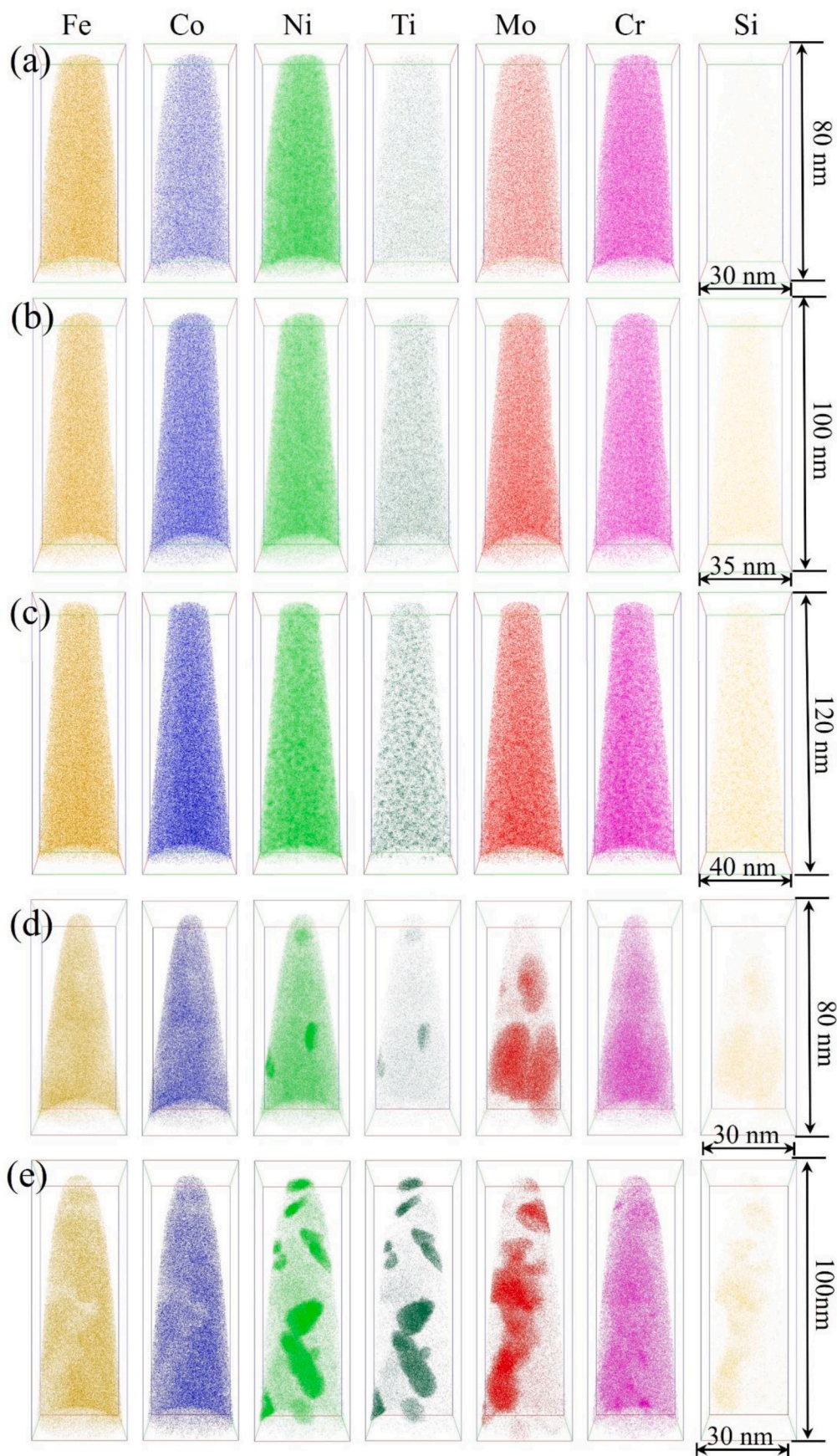
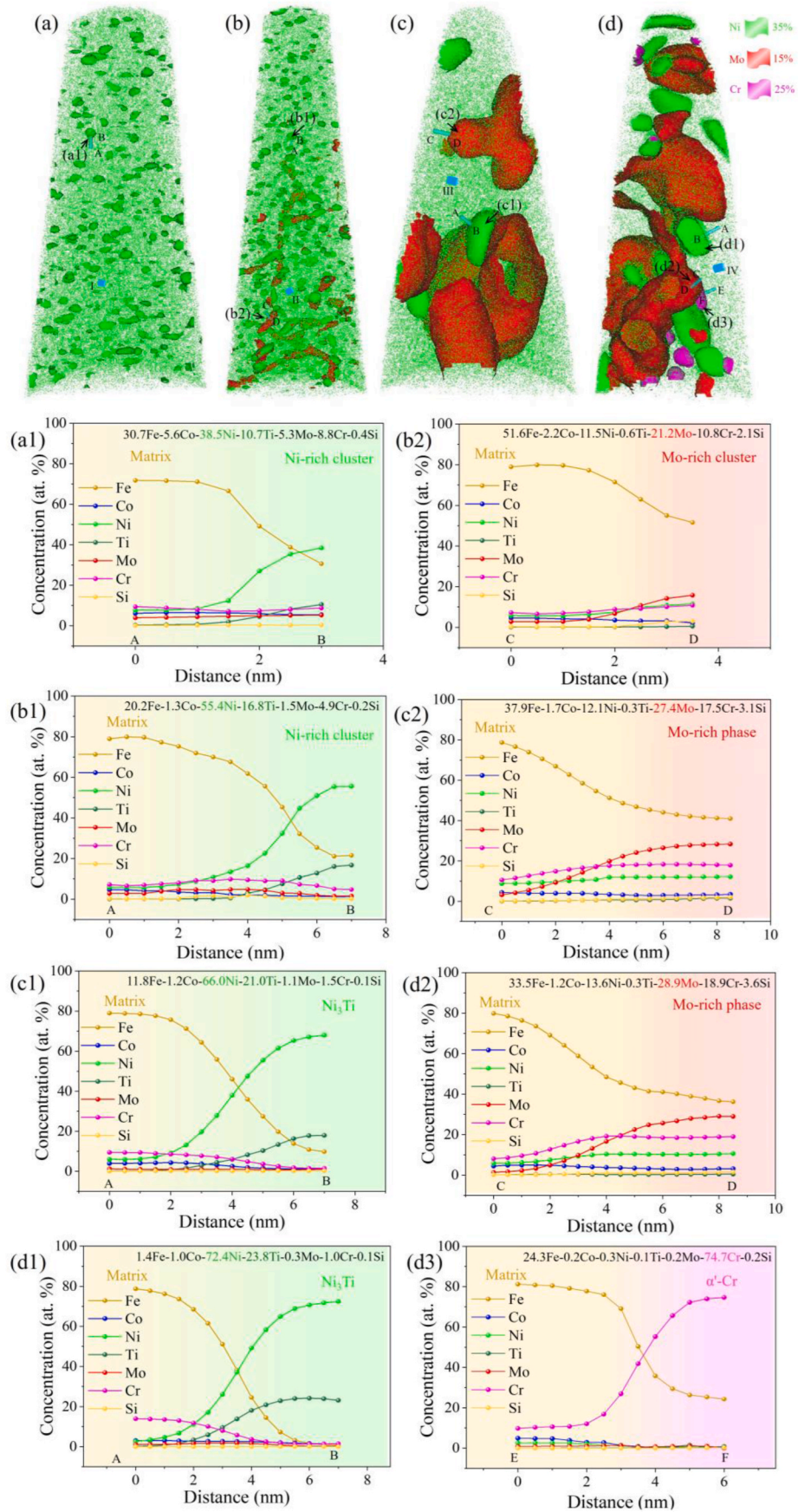


Fig. 6. 3-D atom distribution patterns of (a) ST MSS, (b) SA30 MSS, (c) SA300 MSS, (d) SA3000 MSS and (e) SA30000 MSS.



(caption on next page)

Fig. 7. Typical APT reconstruction of nanoprecipitates in (a) SA30 MSS (b) SA300 MSS (c) SA3000 MSS and (d) SA30000 MSS. Reconstruction iso-concentration interfaces plotted by 35 % Ni (green surfaces), 25 % Cr (pink surfaces) and 15 % Mo (red surfaces). (a1) to (d3) are the corresponding one-dimensional concentration plots via the selected regions marked in a, b, c and d, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 1

Equivalent radii (nm), number densities (10^{23} m^{-3}), and volume fractions of the nanoprecipitates at different aging conditions.

aging time		equivalent radius	number density	volume fraction
30 min	Ni-rich cluster	0.95 ± 0.31	60.56 ± 0.98	3.12 ± 0.91
300 min	Ni-rich cluster	1.11 ± 0.53	50.12 ± 0.97	3.81 ± 1.11
	Mo-rich cluster	1.47 ± 2.15	2.01 ± 0.12	2.57 ± 1.51
3000 min	Ni ₃ Ti	3.43 ± 1.86	4.71 ± 0.89	4.37 ± 0.73
	Mo-rich phase	6.81 ± 1.93	0.71 ± 0.47	8.56 ± 1.13
30,000 min	Ni ₃ Ti	3.89 ± 1.91	5.13 ± 1.27	5.46 ± 1.21
	Mo-rich phase	8.02 ± 2.45	0.53 ± 0.27	11.63 ± 2.27
	α' -Cr	1.53 ± 0.42	1.57 ± 0.37	1.81 ± 0.34

Meanwhile, Mo-rich phase are also formed due to the enrichment of Mo atoms in the matrix. With further extension of aging time, Ni₃Ti becomes stable with no significant size growth. The Mo-rich phase grow, and some precipitates grow sharply to form large-sized spheres at different locations. Meanwhile, the α' -Cr also appear, mainly distributing at the interface between the Ni₃Ti and matrix.

4.2. Strengthening mechanisms

There are three contributors to the yield strength of maraging steels [38]: (1) strength contribution of the lath martensite, σ_{Mart} , (2) solid solution strengthening, σ_{SS} , and (3) precipitation strengthening of some intermetallic compounds, σ_p . In the current MSS, RA is observed after long-time aging (Fig. 3). However, the strength increment remains a high level in spite of the presence of RAs. It is reported that RAs have almost no contribution on the strength of the steel [2,3]. Therefore, the YS σ_Y of the MSS can be expressed as Eq. [3]:

$$\sigma_Y = \sigma_{Mart} + \sigma_{ss} + \sigma_p \quad (4)$$

The strength of the martensite matrix relies on dislocation density and grain dimension. The block size is employed as the effective grain size in martensite. Grain boundary strengthening is represented by the Hall-Petch equation for d_{Block} [38], while the Taylor equation takes into account the effect of the change in the dislocation density on the strengthening contribution. The σ_{Mart} can be expressed by Eq. (5), which includes grain boundary strengthening, σ_{gb} , and dislocation strengthening, σ_d :

$$\sigma_{Mart} = \frac{300}{\sqrt{d_{block}}} + 0.25M\mu b\sqrt{\rho} \quad (5)$$

where $M = 2.5$ is the Taylor orientation factor [16], ρ is the density of dislocations in the martensite matrix, $b = 0.28 \text{ nm}$ is the Burgers vector, and $\mu = 71 \text{ GPa}$ is the shear modulus [15]. The values of d_{Block} and ρ can be obtained from Fig. 2.

The solid solution strengthening σ_{SS} is derived from the Fleischer's in Eq. (6), which estimates the increase of critical resolved shear stress owing to displacing atoms [60].

$$\sigma_{ss} = \sum_i (\beta_i^2 x_{i,\alpha})^{0.5} \quad (6)$$

where $x_{i,\alpha}$ is the proportion of solid solution atoms i in the matrix and β_i

is the strengthening constant that relates to atom i [61,62]. The β_i values are obtained in the previous studies [2], as shown in Table 3. The β_i values of different elements are considerably different. As the aging time increases, the nanoprecipitate forming elements gradually segregate, which causes a decrease of the concentration of nanoprecipitate forming elements in the matrix. This significantly weakens the solid solution strengthening effect. The reduction is derived from the mass equilibrium equation, where the solute concentration $x_{i,\alpha}$ in the matrix is denoted by Eq. (7) [62].

$$x_{i,\alpha} = \frac{x_i - \sum_j x_{ij} f_j}{1 - \sum_j f_j} \quad (7)$$

where f_j is the proportion of nanoprecipitates, j and x_{ij} are the balanced content of atom i in j , respectively. In order to make the measurement simpler, the atom proportion of the substitution atom i in the matrix during aging obtained using the APT results as shown in Fig. 7.

The strengthening of lath martensite σ_{Mart} and solid solution strengthening σ_{SS} are shown in Fig. 9. Both show a decreasing trend with increasing aging time. The σ_{Mart} is reduced owing to a slight increase in grain size and a decrease in dislocation density after long-time aging. The reason for the decrease in σ_{SS} is that the precipitation of nanoprecipitates depletes the nanoprecipitate-forming elements in the matrix. However, the yield strength still increases considerably. This indicates that the strength increment in σ_Y is mainly contributed by precipitation strengthening σ_p .

The strength increment from the nanoprecipitates is contributed either by dislocation shearing or bypassing (Orowan mechanism), which has been extensively described in many previous reports [14,15,18,36,63].

For the shearing mechanism, three primary components contributing to the YS increment are coherency strengthening ($\Delta\sigma_{coherency}$), modulus mismatch strengthening ($\Delta\sigma_{modulus}$) and order strengthening ($\Delta\sigma_{orders}$) [16]. The equations for evaluating the strength increments caused by the dislocation shearing mechanisms are as follows:

$$\Delta\sigma_{coherency} = 4.1MG\epsilon^{3/2}f^{1/2}\left(\frac{r}{b}\right)^{1/2} \quad (8)$$

$$\Delta\sigma_{modulus} = 0.0055M(\Delta G)^{3/2}\left(\frac{2f}{G}\right)^{1/2}\left(\frac{r}{b}\right)^{\frac{3m}{2}} - 1 \quad (9)$$

$$\Delta\sigma_{order} = M\left(\frac{\gamma_{apb}^{3/2}}{b}\right)\left(\frac{4r_s f}{\pi T}\right)^{1/2} \quad (10)$$

where $M = 2.5$ is the Taylor factor for BCC metals in tension [38], $\epsilon = 2|(a_p - a_m)/(a_m + a_p)|$ is the constrained lattice parameter mismatch [40], $m = 0.85$ is a constant [60], r is the precipitate radius (Table 1), $r_s = (2/3)^{1/2}r$ is the mean precipitate radius in the glide plane, and f is the volume fraction of precipitates (Table 1). $\Delta G = |G - G_p|$ is the modulus difference between the precipitates and matrix, where G_p is the shear modulus of precipitates (55 GPa for Ni₃Ti [64], 55 GPa for the Mo-rich phase [65], and 63 GPa for the α' -Cr phase [66]). γ_{apb} is the anti-phase boundary energy of precipitates (0.52 J/m² for Ni₃Ti [67], 0.34 J/m² for the Mo-rich phase [65], and 0.38 J/m² for the α' -Cr phase [66]). $\Gamma = Gb^2/2$ is the line tension of the dislocations [68], and $\nu = 0.3$ is the Poisson ratio [13].

The YS increment ($\Delta\sigma_{Orowan}$) from the Orowan looping mechanism is expressed as [69]:

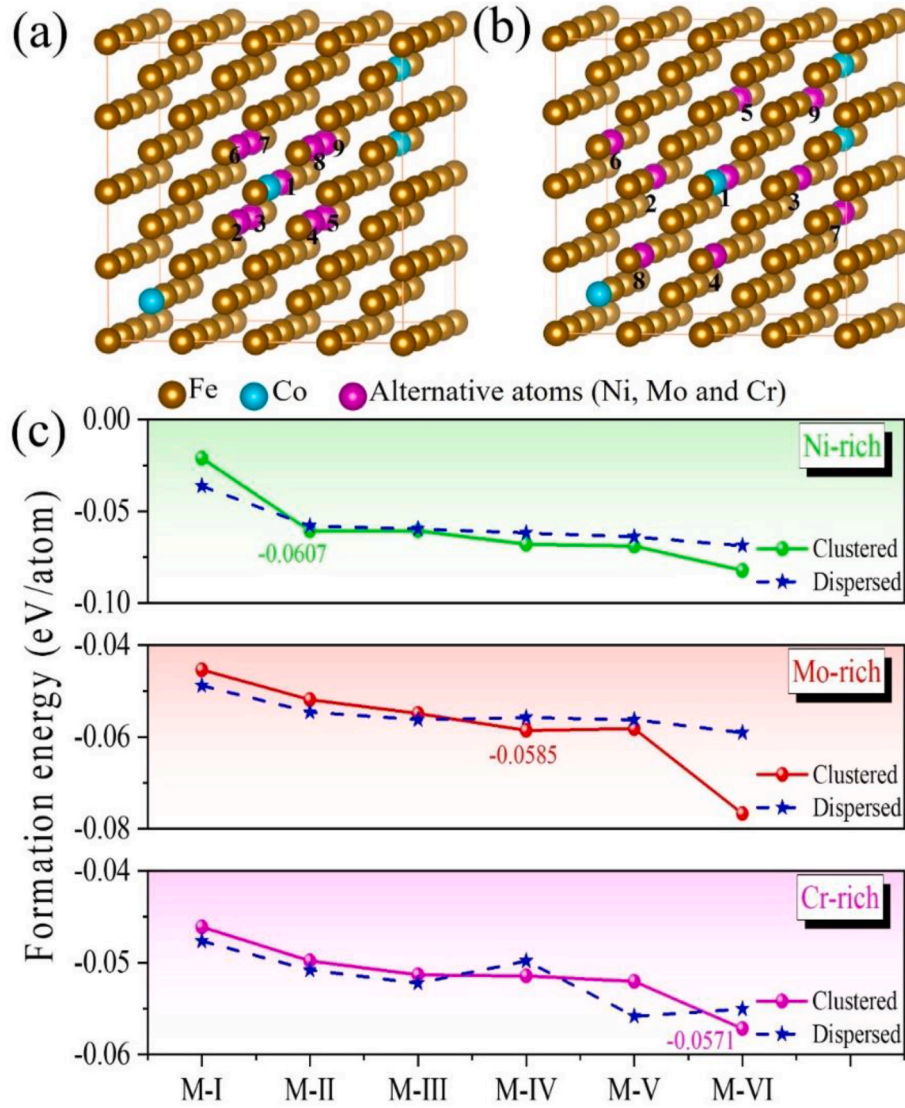


Fig. 8. Atom positions of Co and Fe in (a) System A and (b) System B. Substitutional atoms marked as 1 to 9 replace Fe at the numbered sites. System A was used to evaluate the first nearest neighbors of atoms interactions. System B was used to evaluate configuration with remote atoms positions. (c) Diagram of the formation energy of Ni-rich, Mo-rich and Cr-rich nanoprecipitates at dispersed and clustered distribution conditions.

Table 2
Detailed distribution of atoms for each system employed in the calculation.

Distribution	Model	Systems (Ni atoms as an example, Mo and Cr atoms are arranged in the same way)
Preference of alternative atoms (Ni, Mo or Cr) to be dispersed or clustered with Co	M-I	A (cluster) Fe122Co4Ni2 (Sites 1 and 2)
	M-II	B (dispersed) Fe122Co4Ni2 (Sites 1 and 9)
	M-III	Fe121Co4Ni3 (Sites 1, 2 and 8)
	M-IV	Fe121Co4Ni3 (Sites 1, 4 and 5)
	M-V	Fe123Ni5 (Sites 1 to 5)
	M-VI	Fe123Ni5 (Sites 1, 2, 3, 8 and 9)
		Fe115Co4Ni9 (Sites 1 to 8)

Table 3
Solid solution strengthening constants [2].

Element	Ni	Cr	Ti	Mo	Co	Si
β_i (MPa/at.)	708	622	2628	2362	334	732

$$\Delta\sigma_{\text{Orowan}} = M \frac{0.4Gb}{\pi\sqrt{1-\nu}} \frac{1}{L} \ln\left(\frac{2r_s f}{b}\right) \quad (11)$$

where $L = 2r_s(\pi/4f)^{1/2}$ is the surface-to-surface distance between precipitates.

When more than one species of nanoprecipitates exist, the total strength enhancement can be superimposed by their respective contributions [60]:

$$\sigma_p = \left(\sum_j \Delta\sigma_j^2 \right)^{0.5} \quad (12)$$

where σ_j is the strength increment via a type of nanoprecipitates j with a volume fraction f_j and radius r_j , respectively. The equation is consistent

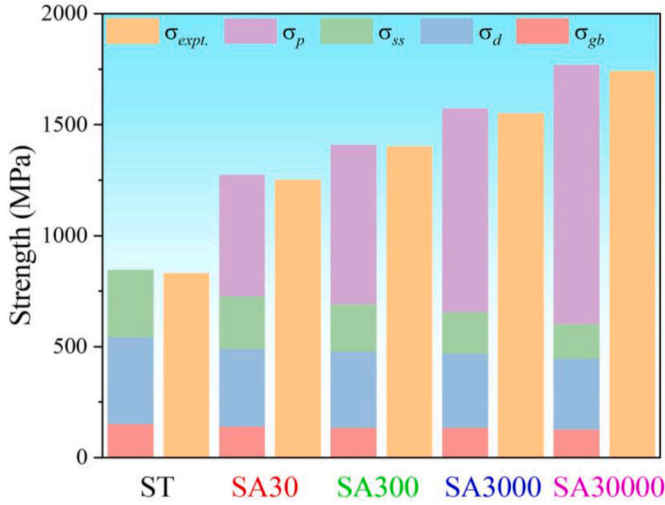


Fig. 9. Total YS of ST, SA300 and SA30000 MSS contributed by each strengthening mechanism.

with the lower strengthening effect produced when different nanoprecipitates form at the interface of other metallic interstitials [70], because their individual contributions are lower than their separate formation.

In the SA30 and SA300 MSS, a large number of fine Ni-rich and Mo-rich cluster were observed in the results of APT. During plastic deformation, additional energy from the anti-phase boundary is demanded by the dislocations to shear these nanoprecipitates, which is the major factor for increasing the strength at the early aging [45]. Therefore, Ni-rich clusters and Mo-rich clusters are considered to employ the shearing mechanism. The theoretical contributions of Ni-Ti and Mo-rich nanoprecipitates to the precipitation strengthening were calculated according to Eqs. 8–10. In the SA30 MSS, $\Delta\sigma_{order}^{Ni-cluster} = 546$ MPa is larger than $\Delta\sigma_{coherency}^{Ni-cluster} = 123$ MPa and $\Delta\sigma_{modulus}^{Ni-cluster} = 71$ MPa, which means that order strengthening ($\Delta\sigma_{order}^{Ni-cluster}$) makes the major contribution in the shearing mechanism. In the SA300 MSS, $\Delta\sigma_{order}^{Ni-cluster} = 643$ MPa is larger than $\Delta\sigma_{coherency}^{Ni-cluster} = 135$ MPa and $\Delta\sigma_{modulus}^{Ni-cluster} = 77$ MPa, which means that order strengthening ($\Delta\sigma_{order}^{Ni-cluster}$) makes the major contribution in the shearing mechanism. As for Mo-rich cluster, $\Delta\sigma_{order}^{Mo-cluster} = 343$ MPa is greater than $\Delta\sigma_{coherency}^{Mo-cluster} = 190$ MPa and $\Delta\sigma_{modulus}^{Mo-cluster} = 72$ MPa, which implies that order strengthening ($\Delta\sigma_{order}^{Mo-cluster}$) is the dominant contribution in the shearing mechanism. The theoretical strength increments ($\sigma_p^{SA30} = 545$ MPa and $\sigma_p^{SA300} = 727$ MPa) of precipitation strengthening are obtained by substituting $\Delta\sigma_{order}^{Ni-cluster}$ and $\Delta\sigma_{order}^{Mo-cluster}$ into Eq. (12). The theoretical strength is close to the value of experimental strength ($\sigma_{experimental}$), indicating that shear strengthening at the under-aged state is the main mechanism of precipitation strengthening, as shown in Fig. 9.

Upon further aging, the nano precipitates grow gradually. In SA3000 MSS, the nanoprecipitates mainly consist of Ni_3Ti and Mo-rich phases. While in SA30000 MSS, three types of nanoprecipitates were observed in the matrix (Figs. 6 and 7), including Ni_3Ti , Mo-rich and α' -Cr phases. The shearing mechanism is considered in the calculation of Ni_3Ti contribution to the theoretical strength increment. Mo-rich phases are incoherent with the matrix. Therefore, only the bypass mechanism is considered. For the α' -Cr, the shearing mechanism is considered [13].

Fig. 9 demonstrates the contributions of the nanoprecipitates to the theoretical strength increment. In SA3000 MSS, the contribution of Ni_3Ti to the strength increment is dominated by the shearing mechanism, and the contribution of ordered strengthening ($\Delta\sigma_{order}^{Ni_3Ti} = 879$ MPa) is much higher than $\Delta\sigma_{coherency}^{Ni_3Ti} = 143$ MPa and $\Delta\sigma_{modulus}^{Ni_3Ti} = 81$ MPa. This implies that the ordered strengthening of Ni_3Ti plays a dominant role in

the shearing mechanism. While the contribution of Mo-rich phase to the strength increment ($\Delta\sigma_{Orowan}^{Mo-rich}$) is 290 MPa when it is dominated by the bypass mechanism. In SA30000 MSS, the contribution of Ni_3Ti to strength increment is dominated by the shearing mechanism, and the contribution of ordered strengthening ($\Delta\sigma_{order}^{Ni_3Ti} = 1061$ MPa) is much higher than the other two strengthening contributions ($\Delta\sigma_{coherency}^{Ni_3Ti} = 163$ MPa and $\Delta\sigma_{modulus}^{Ni_3Ti} = 91$ MPa). This implies that the ordered strengthening of Ni_3Ti plays a dominant role in the shearing mechanism. While the contribution of Mo-rich precipitate to the strength increment ($\Delta\sigma_{Orowan}^{Mo-rich phase}$) is 372 MPa when it is dominated by the bypass mechanism. For α' -Cr, the ordered strengthening ($\Delta\sigma_{order}^{\alpha'-Cr} = 340$ MPa) is much larger than the modulus mismatch strengthening ($\Delta\sigma_{modulus}^{\alpha'-Cr} = 22$ MPa) and coherency strengthening ($\Delta\sigma_{coherency}^{\alpha'-Cr} = 32$ MPa), which means that the order strengthening of α' -Cr plays a dominant role in the shearing mechanism. The theoretical strength increments ($\sigma_p^{SA3000} = 925$ MPa and $\sigma_p^{SA30000} = 1175$ MPa) of precipitation strengthening are obtained by substituting $\Delta\sigma_{order}^{Ni_3Ti}$, $\Delta\sigma_{Orowan}^{Mo-rich}$ and $\Delta\sigma_{order}^{\alpha'-Cr}$ into Eq. (12). The theoretical strengths agree well with the values of experimental strength.

The contributions of the shearing and Orowan looping mechanisms from the various nanoprecipitates to the increment in strength were evaluated. Based on the properties of nanoprecipitates from TEM and APT, for each kind of precipitate, both shearing and Orowan looping mechanisms are evaluated and compared to the experimental data. Table 4 demonstrates the strengthening contributions from the various nanoprecipitates and the corresponding strengthening mechanism. The Ni-rich clusters and Ni_3Ti precipitates mainly contribute to the strength with the shearing mechanism. Mo-rich clusters contribute to the strength by the shearing mechanism. However, when the Mo-rich clusters coarsen into the hexagonal Mo-rich phase, the strengthening mechanism transfers from shearing mechanism to Orowan looping mechanism. The α' -Cr precipitates contributes the strength mainly by shearing mechanism.

To verify the validity of the strengthening mechanism for various precipitates, the concept of the critical radius was applied. The critical radius is estimated when the strength increments produced by the shearing and Orowan mechanisms are equal [13,71]. Beyond this critical radius, dislocations will bypass precipitates rather than shearing. For Ni_3Ti precipitates, the critical radius is 5.37 nm. The critical radius of Mo-rich precipitates is calculated as 6.52 nm. The critical radius of α' -Cr precipitate is 5.94 nm. The APT results indicate that the radius of Ni_3Ti precipitates is 0.95 nm, 1.11 nm, 3.43 nm and 3.89 nm after aging for 30 min, 300 min, 3000 min and 30000 min, respectively, smaller than the critical radius. Thus, it is reasonable that Ni_3Ti precipitates strengthens the steel by shearing mechanism. The radius of Mo-rich clusters is 1.47 nm, which is smaller than the critical radius. So, the shearing mechanism is reasonable. The size of Mo-rich phases developed from Mo-rich clusters aging for 3000 min and 30000 min are 6.81 nm and 8.02 nm respectively, which are larger than the critical radius. Thus, the bypassing mechanism is reasonable. For α' -Cr precipitates, the size is 1.53 nm, which is smaller than the critical radius. So, the shearing mechanism is reasonable.

4.3. The effects of RA

Typically, the TRIP effect of RA is used to improve ductility in maraging steels [22,72]. In addition, the martensite formation process relaxes stresses by forming martensite-favorable variants in the region of stress concentration, which retards the formation and propagation of cracks and contributes significantly to the improvement of uniform deformation [73]. Comparing the engineering stress-strain curves (Fig. 1 a), the curve of the over-aged state exhibits high strengths without a significant loss of plasticity. This is mainly due to the existence of RA in the over-aged condition, and the RA undergoes a TRIP effect during

Table 4
Strength contribution of different strengthening mechanisms.

Specimens	Precipitate types	σ_{ss}	σ_{gb}	σ_d	Ni-rich cluster/ Ni_3Ti				Mo-rich cluster/phase				α' -Cr		σ_p	σ_{cal}	$\sigma_{expt.}$	
					shearing		modulus	order	shearing		modulus	order	bypass	shearing				coherency
					coherency	modulus			coherency	modulus								
ST	None	301	152	391												844	830	
SA30	Ni-rich cluster	235	141	351	123	71		546							546	1273	1250	
SA300	Ni-rich cluster	213	135	343	135	77		643	190	72		343			727	1418	1400	
SA3000	Mo-rich cluster																	
	Ni_3Ti	186	135	335	143	81		879					290		925	1581	1550	
SA30000	Mo-rich phase																	
	Ni_3Ti	156	127	320	163	91		1061					372	22	1175	1778	1750	
	Mo-rich phase																	
	α' -Cr																	

deformation. The RAs in the present work do not contain nano-precipitates (Fig. 4). Although the MSS can be ensured to obtain ductility with ultra-high strength. However, the work-hardening ability was not significantly improved (Fig. 1).

5. Conclusions

The high strength MSS was developed utilizing multiple nano-precipitates and RA. The steel shows an excellent combination of mechanical properties with a YS of ~1750 MPa, a TS of ~1910 MPa and a TE of ~10.5 % at room temperature. The following conclusions can be drawn.

1. At different aging time at 480 °C, the matrix consists of martensite with a high dislocation density. Three types of nanoprecipitates and RA appear in the matrix after aging, including Ni_3Ti , Mo-rich, and α' -Cr precipitates. The contribution of strength increment comes from precipitation strengthening. Ductility increases from the TRIP effect of RA.
2. During aging, there is a certain precipitation sequence: Ni-rich cluster→ Ni-rich cluster + Mo-rich cluster→ Ni_3Ti + Mo-rich phase→ Ni_3Ti + Mo-rich phase + α' -Cr. The evolution of the 0.95 nm Ni-rich clusters into 3.43 nm Ni_3Ti is not significantly coarsened. Because the Mo-rich phase wrapped around Ni-rich cluster or Ni_3Ti that inhibit the coarsening of Ni_3Ti to ensure a strong precipitation strengthening effect after long-time aging. Conversely, Mo-rich clusters coarsen significantly from 1.47 nm to 6.81 nm of Mo-rich phase, which leads to a shift in the precipitation strengthening mechanism. The radius of the finally precipitated α' -Cr is 1.53 nm.
3. In the early stage of aging, only Ni-rich clusters are present, while Mo-rich and α' -Cr phases gradually appear with the increasing aging time. The concentrations of Ni and Ti in Ni-rich cluster cores gradually increase, while the concentrations of Fe, Co, Mo and Cr gradually decrease. Finally, the ratio of Ni to Ti in the cluster cores approaches 3:1 and reaches a stable state, forming the hexagonal structure Ni_3Ti . The Ni-rich cluster formed at the beginning of aging can serve as heterogeneous nucleation sites for the Mo-rich and α' -Cr phase.
4. The yield strength increment increases with the increase of the nanoprecipitate types. The Ni-rich clusters offer precipitation strengthening contributions of 545 MPa. Ni-rich clusters together with Mo-rich clusters provide 720 MPa precipitation strengthening contribution. Ni_3Ti coupled with coarsened Mo-rich phase gives 920 MPa precipitation strengthening contribution. Ni_3Ti combined with Mo-rich phase and α' -Cr provide 1170 MPa precipitation strengthening contribution. Ni-rich clusters, Mo-rich clusters, Ni_3Ti and α' -Cr contribute to precipitation strengthening mainly by shear mechanisms, where ordered strengthening being the most significant. The Mo-rich phases contribute to the precipitation strengthening primarily through a bypass mechanism.

CRediT authorship contribution statement

Junpeng Li: Writing – review & editing, Writing – original draft, Formal analysis, Data curation. **Weiguo Jiang:** Validation. **Yang Zhang:** Writing – review & editing, Methodology. **Liuyan Liu:** Writing – review & editing, Writing – original draft, Investigation. **Yongzheng Yu:** Writing – original draft, Methodology. **Junhua Luan:** Visualization, Validation, Methodology. **Zengbao Jiao:** Writing – review & editing, Validation, Supervision. **Chain Tsuan Liu:** Writing – review & editing, Supervision. **Zhongwu Zhang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- [1] K.H. Lo, C.H. Shek, J. Lai, Recent developments in stainless steels, *Mater. Sci. Eng. R* 65 (4–6) (2009) 39–104.
- [2] E. Galindo-Nava, W. Rainforth, P. Rivera-Díaz-del-Castillo, Predicting microstructure and strength of maraging steels: elemental optimisation, *Acta Mater.* 117 (2016) 270–285.
- [3] E. Galindo-Nava, P. Rivera-Díaz-del-Castillo, Understanding the factors controlling the hardness in martensitic steels, *Scripta Mater.* 110 (2016) 96–100.
- [4] S. Xu, J. Li, Y. Cui, Y. Zhang, L. Sun, J. Li, J. Luan, Z. Jiao, X.-L. Wang, C. Liu, Mechanical properties and deformation mechanisms of a novel austenite-martensite dual phase steel, *Int. J. Plast.* 128 (2020) 102677.
- [5] M. Niu, K. Yang, J. Luan, W. Wang, Z. Jiao, Cu-assisted austenite reversion and enhanced TRIP effect in maraging stainless steels, *J. Mater. Sci. Technol.* 104 (2022) 52–58.
- [6] H. Zhang, M. Sun, Y. Liu, D. Ma, B. Xu, M. Huang, D. Li, Y. Li, Ultrafine-grained dual-phase maraging steel with high strength and excellent cryogenic toughness, *Acta Mater.* 211 (2021) 116878.
- [7] S.H. Nedjad, M.N. Ahmadabadi, T. Furuhaara, Correlation between the intergranular brittleness and precipitation reactions during isothermal aging of an Fe–Ni–Mn maraging steel, *Mater. Sci. Eng., A* 490 (1–2) (2008) 105–112.
- [8] N. Heo, Ductile-brittle-ductile transition and grain boundary segregation of Mn and Ni in an Fe–6Mn–12Ni alloy, *Scripta Mater.* 34 (10) (1996) 1517–1522.
- [9] W. Xu, P. Rivera-Díaz-del-Castillo, W. Yan, K. Yang, D. San Martín, L. Kestens, V. D. Zwaag, A new ultrahigh-strength stainless steel strengthened by various coexisting nanoprecipitates, *Acta Mater.* 58 (11) (2010) 4067–4075.
- [10] Y. Li, W. Yan, J.D. Cotton, G.J. Ryan, Y. Shen, W. Wang, Y. Shan, K. Yang, A new 1.9 GPa maraging stainless steel strengthened by multiple precipitating species, *Mater. Des.* 82 (2015) 56–63.
- [11] K. Stiller, F. Danoix, A. Bostel, Investigation of precipitation in a new maraging stainless steel, *Appl. Surf. Sci.* 94 (1996) 326–333.
- [12] J. Tian, M.B. Shahzad, W. Wang, L. Yin, Z. Jiang, K. Yang, Role of Co in formation of Ni–Ti clusters in maraging stainless steel, *J. Mater. Sci. Technol.* 34 (9) (2018) 1671–1675.
- [13] M. Niu, L. Yin, K. Yang, J. Luan, W. Wang, Z. Jiao, Synergistic alloying effects on nanoscale precipitation and mechanical properties of ultrahigh-strength steels strengthened by Ni₃Ti, Mo-enriched, and Cr-rich co-precipitates, *Acta Mater.* 209 (2021) 116788.
- [14] M. Niu, G. Zhou, W. Wang, M.B. Shahzad, Y. Shan, K. Yang, Precipitate evolution and strengthening behavior during aging process in a 2.5 GPa grade maraging steel, *Acta Mater.* 179 (2019) 296–307.
- [15] Z. Wang, B. Niu, Q. Wang, C. Dong, J. Jie, T. Wang, T. Nieh, Designing ultrastrong maraging stainless steels with improved uniform plastic strain via controlled precipitation of coherent nanoparticles, *J. Mater. Sci. Technol.* 93 (2021) 60–70.
- [16] L. Sun, T. Simm, T. Martin, S. McAdam, D. Galvin, K. Perkins, P. Bagot, M. Moody, S. Ooi, P. Hill, A novel ultra-high strength maraging steel with balanced ductility and creep resistance achieved by nanoscale β -NiAl and Laves phase precipitates, *Acta Mater.* 149 (2018) 285–301.
- [17] Z. Jiao, J. Luan, M. Miller, Y. Chung, C. Liu, Co-precipitation of nanoscale particles in steels with ultra-high strength for a new era, *Mater. Today* 20 (3) (2017) 142–154.
- [18] S. Xu, Y. Zhao, D. Chen, L. Sun, L. Chen, X. Tong, C. Liu, Z. Zhang, Nanoscale precipitation and its influence on strengthening mechanisms in an ultra-high strength low-carbon steel, *Int. J. Plast.* 113 (2019) 99–110.
- [19] S. Jin, X. Cao, G. Cheng, X. Lian, T. Zhu, P. Zhang, R. Yu, B. Wang, Thermally promoted evolution of open-volume defects and Cu precipitates in the deformed FeCu alloys, *J. Nucl. Mater.* 501 (2018) 293–301.
- [20] Z. Jiao, J. Luan, M. Miller, C. Yu, Y. Liu, C.J. Liu, Precipitate transformation from NiAl-type to Ni₂AlMn-type and its influence on the mechanical properties of high-strength steels, *Acta Mater.* 110 (2016) 31–43.
- [21] S.-H. Kim, H. Kim, N.J. Kim, Brittle intermetallic compound makes ultrastrong low-density steel with large ductility, *Nature* 518 (7537) (2015) 77–79.
- [22] B. He, B. Hu, H. Yen, G. Cheng, Z. Wang, H. Luo, M. Huang, High dislocation density-induced large ductility in deformed and partitioned steels, *Science* 357 (6355) (2017) 1029–1032.
- [23] J. Gao, S. Jiang, H. Zhang, Y. Huang, D. Guan, Y. Xu, S. Guan, L.A. Bendersky, A. V. Davydov, Y. Wu, Facile route to bulk ultrafine-grain steels for high strength and ductility, *Nature* 590 (7845) (2021) 262–267.
- [24] L. Liu, Y. Zhang, J. Ma, Y. Shang, J. Han, J. Li, Z. Zhang, Transition of plastic deformation mechanisms governed by spacing of nano-precipitates in a high entropy alloy, *Scripta Mater.* 217 (2022) 114771.
- [25] J. Hu, L.-X. Du, G.-S. Sun, H. Xie, R. Misra, The determining role of reversed austenite in enhancing toughness of a novel ultra-low carbon medium manganese high strength steel, *Scripta Mater.* 104 (2015) 87–90.
- [26] L. Yuan, D. Ponge, J. Wittig, P. Choi, J.A. Jiménez, D. Raabe, Nanoscale austenite reversion through partitioning, segregation and kinetic freezing: example of a ductile 2 GPa Fe–Cr–C steel, *Acta Mater.* 60 (6–7) (2012) 2790–2804.
- [27] Y. Fan, B. Zhang, H. Yi, G. Hao, Y. Sun, J. Wang, E.-H. Han, W. Ke, The role of reversed austenite in hydrogen embrittlement fracture of S41500 martensitic stainless steel, *Acta Mater.* 139 (2017) 188–195.
- [28] D. Raabe, S. Sandlöbes, J. Millán, D. Ponge, H. Assadi, M. Herbig, P.-P. Choi, Segregation engineering enables nanoscale martensite to austenite phase transformation at grain boundaries: a pathway to ductile martensite, *Acta Mater.* 61 (16) (2013) 6132–6152.
- [29] J. Escobar, G. Faria, E. Maia, J. Oliveira, T. Boll, S. Seils, P. Mei, A. Ramirez, Fundamentals of isothermal austenite reversion in a Ti-stabilized 12Cr–6 Ni–2 Mo super martensitic stainless steel: thermodynamics versus experimental assessments, *Acta Mater.* 174 (2019) 246–259.
- [30] L. Huang, W. Lin, K. Wang, S. Song, C. Guo, Y. Chen, Y. Li, F. Liu, Grain boundary-constrained reverse austenite transformation in nanostructured Fe alloy: model and application, *Acta Mater.* 154 (2018) 56–70.
- [31] Y.-U. Heo, M. Takeguchi, K. Furuya, H.-C. Lee, Transformation of DO24 η -Ni₃Ti phase to face-centered cubic austenite during isothermal aging of an Fe–Ni–Ti alloy, *Acta Mater.* 57 (4) (2009) 1176–1187.
- [32] G.T. Williamson, R. Smallman, The use of Fourier analysis in the interpretation of X-ray line broadening from cold-worked iron and molybdenum, *Acta Crystallogr.* 7 (8–9) (1954) 574–581.
- [33] T. Liu, Z. Cao, H. Wang, G. Wu, J. Jin, W. Cao, A new 2.4 GPa extra-high strength steel with good ductility and high toughness designed by synergistic strengthening of nano-particles and high-density dislocations, *Scripta Mater.* 178 (2020) 285–289.
- [34] L. Liu, Y. Zhang, J. Li, M. Fan, X. Wang, G. Wu, Z. Yang, J. Luan, Z. Jiao, C.T. Liu, Enhanced strength-ductility synergy via novel bifunctional nano-precipitates in a high-entropy alloy, *Int. J. Plast.* 153 (2022) 103235.
- [35] Y. Zhao, X. Tong, X. Wei, S. Xu, S. Lan, X.-L. Wang, Z. Zhang, Effects of microstructure on crack resistance and low-temperature toughness of ultra-low carbon high strength steel, *Int. J. Plast.* 116 (2019) 203–215.
- [36] X. Wei, X. Cao, J. Luan, Z. Jiao, C. Liu, Z. Zhang, Synergy of strengthening and toughening of a Cu-rich precipitate-strengthened steel, *Mater. Sci. Eng. A* 832 (2022) 142487.
- [37] Y. Song, X. Li, L. Rong, Y. Li, The influence of tempering temperature on the reversed austenite formation and tensile properties in Fe–13% Cr–4% Ni–Mo low carbon martensite stainless steels, *Mater. Sci. Eng., A* 528 (12) (2011) 4075–4079.
- [38] E. Galindo-Nava, P. Rivera-Díaz-del-Castillo, A model for the microstructure behaviour and strength evolution in lath martensite, *Acta Mater.* 98 (2015) 81–93.
- [39] K. Macek, P. Lukáš, J. Janovec, P. Mikula, P. Strunz, M. Vrána, M. Zaffagnini, Austenite content and dislocation density in electron-beam welds of a stainless maraging steel, *Mater. Sci. Eng., A* 208 (1) (1996) 131–138.
- [40] S. Jiang, H. Wang, Y. Wu, X. Liu, H. Chen, M. Yao, B. Gault, D. Ponge, D. Raabe, A. Hirata, Ultrastrong steel via minimal lattice misfit and high-density nanoprecipitation, *Nature* 544 (7651) (2017) 460–464.
- [41] C.-Y. Huang, H.-W. Yen, HRTEM investigations on nano precipitates in Custom 475 maraging stainless steel, *Mater. Char.* 178 (2021) 111216.
- [42] J. Tian, W. Wang, L. Yin, W. Yan, Y. Shan, K. Yang, Three dimensional atom probe and first-principles studies on spinodal decomposition of Cr in a Co-alloyed maraging stainless steel, *Scripta Mater.* 121 (2016) 37–41.
- [43] K. Stiller, F. Danoix, M. Hättestrand, Mo precipitation in a 12Cr–9Ni–4Mo–2Cu maraging steel, *Mater. Sci. Eng., A* 250 (1) (1998) 22–26.
- [44] Z. Guo, W. Sha, D. Vaumousse, Microstructural evolution in a PH13-8 stainless steel after ageing, *Acta Mater.* 51 (1) (2003) 101–116.
- [45] H. Leitner, M. Schöber, R. Schnitzer, Splitting phenomenon in the precipitation evolution in an Fe–Ni–Al–Ti–Cr stainless steel, *Acta Mater.* 58 (4) (2010) 1261–1269.
- [46] E.V. Pereloma, A. Shekhter, M.K. Miller, S.P. Ringer, Ageing behaviour of an Fe–20Ni–1.8 Mn–1.6 Ti–0.59 Al (wt%) maraging alloy: clustering, precipitation and hardening, *Acta Mater.* 52 (19) (2004) 5589–5602.
- [47] Z.X. A, I.T. B, E.P.C. D, Clustering, nano-scale precipitation and strengthening of steels, *Prog. Mater. Sci.* (2020).
- [48] M. Hättestrand, J.-O. Nilsson, K. Stiller, P. Liu, M. Andersson, Precipitation hardening in a 12% Cr–9% Ni–4% Mo–2% Cu stainless steel, *Acta Mater.* 52 (4) (2004) 1023–1037.
- [49] H. Nitta, T. Yamamoto, R. Kanno, K. Takasawa, T. Iida, Y. Yamazaki, S. Ogu, Y. Iijima, Diffusion of molybdenum in α -iron, *Acta Mater.* 50 (16) (2002) 4117–4125.
- [50] G. Hettich, H. Mehrer, K. Maier, Self-diffusion in ferromagnetic α -iron, *Scripta Mater.* 11 (9) (1977) 795–802.

- [51] V. Shapovalov, A. Kurasov, Diffusion of titanium in iron, *Met. Sci. Heat Treat.* 17 (9) (1975) 803–805.
- [52] R. Braun, M. Feller-Kniepmeier, Diffusion of chromium in α -iron, *Phys. Status Solodi* 90 (2) (1985) 553–561.
- [53] X. Li, Z. Yin, Y. Wang, M. Zheng, Mössbauer study of the composition fluctuation during spinodal decomposition in a 3.5 GPa Fe-10% Ni-14% Mo-19% Co (wt.%) maraging steel, *Mater. Sci. Eng., A* 212 (1) (1996) 182–185.
- [54] H. Hou, H. Li, Y. Jin, X. Wang, Z. Wen, Effect of heat treatment temperature on the mechanical properties of low-temperature high strength maraging steel, *Mater. Sci. Eng., A* 601 (2014) 1–6.
- [55] C. Zhang, C. Wang, S. Zhang, Y. Ding, Q. Ge, J. Su, Effect of aging temperature on the precipitation behavior and mechanical properties of Fe–Cr–Ni maraging stainless steel, *Mater. Sci. Eng., A* 806 (2021) 140763.
- [56] R. Tewari, S. Mazumder, I. Batra, G. Dey, S. Banerjee, Precipitation in 18 wt% Ni maraging steel of grade 350, *Acta Mater.* 48 (5) (2000) 1187–1200.
- [57] A.S. Murthy, J.E. Medvedeva, D. Isheim, S.L. Lekakh, V.L. Richards, D.C. Van Aken, Copper precipitation in cobalt-alloyed precipitation-hardened stainless steel, *Scripta Mater.* 66 (11) (2012) 943–946.
- [58] S. Baker, A. Asaduzzaman, M. Roy, S. Gurman, C. Binns, J. Blackman, Y. Xie, Atomic structure and magnetic moments in cluster-assembled nanocomposite Fe/Cu films, *Phys. Rev. B Condens. Matter* 78 (1) (2008) 014422.
- [59] L. Kong, B. Liu, Correlation of magnetic moment versus spacing distance of metastable fcc structured iron, *Appl. Phys. Lett.* 84 (18) (2004) 3627–3629.
- [60] A.J. Ardell, Precipitation hardening, *Metall. Trans. A* 16 (12) (1985) 2131–2165.
- [61] R.L. Fleischer, Substitutional solution hardening, *Acta Mater.* 11 (3) (1963) 203–209.
- [62] R. Labusch, A statistical theory of solid solution hardening, *Phys. Status Solodi* 41 (2) (1970) 659–669.
- [63] L. Liu, Y. Zhang, J. Han, X. Wang, W. Jiang, C.T. Liu, Z. Zhang, P.K. Liaw, Nanoprecipitate-Strengthened high-entropy alloys, *Adv. Sci.* 8 (23) (2021) 2100870.
- [64] Q. Chen, Z. Huang, Z. Zhao, C. Hu, First-principles study on the structural, elastic, and thermodynamics properties of Ni3X (X: Al, Mo, Ti, Pt, Si, Nb, V, and Zr) intermetallic compounds, *Appl. Phys. A* 116 (3) (2014) 1161–1172.
- [65] K.S. Chan, Alloying effects on the fracture toughness of Nb-based silicides and Laves phases, *Mater. Sci. Eng., A* 409 (1–2) (2005) 257–269.
- [66] D. Schwen, E. Martinez, A. Caro, On the analytic calculation of critical size for alpha prime precipitation in FeCr, *J. Nucl. Mater.* 439 (1–3) (2013) 180–184.
- [67] J. Wang, H. Sehitoglu, Dislocation slip and twinning in Ni-based L12 type alloys, *Intermetallics* 52 (2014) 20–31.
- [68] A. Kelly, R. Nicholson. *Strengthening Methods in Crystals*, Elsevier publishing company, 1971.
- [69] K. Ma, H. Wen, T. Hu, T.D. Topping, D. Isheim, D.N. Seidman, E.J. Lavernia, J. M. Schoenung, Mechanical behavior and strengthening mechanisms in ultrafine grain precipitation-strengthened aluminum alloy, *Acta Mater.* 62 (2014) 141–155.
- [70] M. Kapoor, D. Isheim, G. Ghosh, S. Vaynman, M.E. Fine, Y.-W. Chung, Aging characteristics and mechanical properties of 1600 MPa body-centered cubic Cu and B2-NiAl precipitation-strengthened ferritic steel, *Acta Mater.* 73 (2014) 56–74.
- [71] J. Lee, H. Kim, K. Jeong, S.-J. Park, J. Moon, S.-g. Kang, H.N. Han, Prediction of precipitation kinetics and strengthening in FeMnAlC lightweight steels, *J. Mater. Res. Technol.* 14 (2021) 2897–2908.
- [72] M. Wang, C.C. Tasan, D. Ponge, A. Kostka, D. Raabe, Smaller is less stable: size effects on twinning vs. transformation of reverted austenite in TRIP-maraging steels, *Acta Mater.* 79 (2014) 268–281.
- [73] X. Chen, L. Zheng, Q. Zhu, S. Zhao, F. Yu, M. Liu, H. Liu, H. Zhang, H. Xu, Nanoscale interface engineering in a high Cr-Co martensitic steel, *Acta Mater.* 271 (2024) 119901.